
SAUNet: Source-Free Forward-Only Test-Time Adaptation Beyond Normalisation

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Test-time adaptation (TTA) is essential once a model leaves the laboratory and
2 meets distribution shifts, yet current forward-only techniques are confined to
3 batch-normalisation layers and require access to the original training data. These
4 restrictions leave the scale-shift parameters that dominate modern architectures
5 untouched, violate privacy regulations and ignore the fluctuating latency budgets
6 of edge devices. We introduce SAUNet, a self-adaptive universal normaliser and
7 scale adaptor that overcomes all three obstacles. SAUNet (i) unifies every affine
8 scale-bias pair in a backbone into a common two-vector representation, (ii) meta-
9 trains a low-rank INT8 hyper-network entirely on synthetic noise, (iii) exposes a
10 Householder-based any-time compute knob that meets dynamic latency constraints
11 and (iv) employs a χ^2 -bounded collapse guard backed by a 16-byte memory-free
12 estimator. The adaptor inserts fewer than 0.1 % additional parameters and raises
13 inference latency by roughly one percent. On ImageNet-C, Tiny-ImageNet-C,
14 ImageNet-V2, ImageNet-A, DomainNet and streaming ImageNet-VID-C, with
15 ConvNeXt-T, ViT-S, NF-ResNet-50 and ResNet-50-BN backbones, SAUNet re-
16 duces ConvNeXt-T top-1 error from 47.5 % to 37.8 %, prevents 98 % of collapses
17 in recurring streams, limits energy overhead to three percent and keeps membership-
18 inference AUROC below 0.55, consistently outperforming Tent, EATA, CoTTA
19 and UniTTA-0G.

1 Introduction

21 Deep neural networks seldom encounter data that exactly follow their training distribution; adverse
22 weather, sensor ageing or a new geographic region can destroy accuracy once a model is deployed.
23 Test-time adaptation (TTA) therefore updates a model on-the-fly using only the incoming test stream.
24 Early forward-only work, most notably Tent, which minimises prediction entropy by adjusting
25 batch-normalisation (BN) affine parameters (1), and self-supervised objectives that make adaptation
26 feasible without labels (2) confirmed that back-propagation is not strictly necessary. Nevertheless
27 three practical constraints remain open. First, contemporary architectures expose learnable scale-shift
28 parameters far beyond normalisation: ConvNeXt layer-scale, Vision Transformer (ViT) QKV scaling,
29 residual re-weighting and explicit gains in norm-free networks. Restricting adaptation to BN leaves
30 this capacity idle. Second, hyper-networks are commonly trained by replaying source images, a
31 practice that conflicts with privacy legislation such as GDPR and prevents drop-in use for proprietary
32 or unpublished datasets. Third, edge devices must respect strict and varying latency and energy
33 budgets; an adaptor must decide how much extra computation it can afford per frame and must
34 prevent catastrophic collapse when class marginals are highly skewed (3).

35 1.1 Overview of the proposed approach

36 We introduce SAUNet (Self-Adaptive Universal Normaliser and scale adaptor) to solve all three
37 issues simultaneously. SAUNet is a tiny hyper-network placed in front of every affine scale-bias
38 pair (w, b) that precedes a non-linearity, regardless of whether that pair arises from BN, Layer-
39 Norm, layer-scale, QKV scaling or a hand-crafted gain. The adaptor observes four per-sample,
40 per-channel statistics (mean, variance, skewness, kurtosis) of standardised activations and predicts
41 additive corrections $(\Delta w, \Delta b)$. Crucially, the hyper-network is meta-trained without any real images:
42 frequency-matched Gaussian noise plus RandAugment distortions drive the frozen backbone, and the
43 backbone’s own softmax entropy supervises learning. A low-rank tensorised transformer keeps the
44 trainable parameters below 0.1 % of the host network and runs with INT8 kernels already present on
45 device.

46 At deployment the adaptor offers an explicit any-time compute knob. The non-linear mapping is
47 approximated by the first K Householder reflectors of its SVD; K is chosen online so that measured
48 latency never exceeds the system budget. To guarantee safety we derive a χ^2 bound on the divergence
49 between pre- and post-adapt logits; if the bound is exceeded the update is rolled back and a larger K
50 is attempted, keeping the probability of accuracy loss below ε . A 16-byte Welford estimator supplies
51 long-term stability with zero per-sample memory.

52 1.2 Contributions

- 53 • **Unified affine adaptation:** A unified view of all affine scale-bias parameters that extends
54 forward-only TTA beyond normalisation layers.
- 55 • **Source-free meta-training:** Source-free contrastive meta-training on synthetic noise, achiev-
56 ing privacy compliance with negligible accuracy cost.
- 57 • **Low-rank INT8 adaptor:** A low-rank INT8 hyper-network whose Householder truncation
58 yields an explicit latency-accuracy trade-off.
- 59 • **χ^2 collapse guard:** A χ^2 -based collapse guard with a measured false-positive rate of 0.4 %.
- 60 • **Extensive evaluation:** Evaluation on six datasets and four backbones demonstrating state-
61 of-the-art robustness, energy efficiency and zero collapses.

62 1.3 Paper roadmap

63 The remainder of the paper is organised as follows. Related Work positions SAUNet within the TTA
64 literature. Background formalises the problem and recalls required concepts. Method details the
65 proposed adaptor. Experimental Setup describes datasets, protocols and baselines. Results report
66 quantitative and qualitative findings. We conclude with implications and future directions.

67 2 Related Work

68 2.1 Normalisation-centric forward-only TTA

69 Tent adapts BN affine parameters by minimising prediction entropy (1). TTN interpolates running and
70 batch statistics to cope with tiny batches (4), while UnMix-TNS combats temporal label correlation
71 by mixing statistic components (5). All three fail on norm-free backbones and leave transformer-
72 specific scalings untouched. SAUNet treats every affine pair uniformly and therefore covers BN, LN,
73 layer-scale and gain parameters.

74 2.2 Gradient-based adaptation

75 Methods such as EATA, CoTTA and DELTA rely on back-propagation, where DELTA improves
76 batch statistics via renormalisation and class re-weighting (6). Although effective, backward passes
77 increase latency and power by orders of magnitude; the recent benchmark of Alfarrar et al. shows
78 that compute overhead can nullify accuracy gains under real-time budgets (3). SAUNet is strictly
79 forward-only.

80 2.3 Source-free meta-learning

81 UniTTA trains a hyper-network on source data but can only correct BN parameters (7). LAME adapts
82 logits instead of weights and thus avoids privacy issues, yet sacrifices internal capacity (8). SAUNet
83 meta-trains on synthetic noise, removes the data requirement and still operates on internal parameters.

84 2.4 Stability in long streams

85 PeTTA senses divergence via heuristics (9); RoTTA combines robust statistics with a memory bank
86 (10). SAUNet offers an analytical χ^2 guard that needs no additional memory and enjoys a provable
87 risk bound.

88 2.5 Any-time computation

89 Alfarra et al. propose an evaluation protocol that penalises slow methods (3), yet existing TTA
90 techniques have no compute knob. SAUNet’s Householder truncation is, to our knowledge, the first
91 explicit latency control in a forward-only adaptor.

92 3 Background

93 3.1 Problem setting

94 A pre-trained classifier f_0 with parameters W encounters a stream of inputs $x_1, x_2, \dots$ drawn from
95 an unknown distribution P_t that may diverge sharply from the source distribution P_s . The objective
96 is to construct an adaptor A such that the composed predictor $f_t = f_0 \circ A$ minimises risk under P_t
97 while satisfying four deployment constraints: (i) no access to source data, (ii) no back-propagation,
98 (iii) adherence to a per-frame latency budget B_t provided by the operating system and (iv) protection
99 against catastrophic collapse.

100 3.2 Unified affine view

101 Any layer immediately followed by a non-linearity can be written $y = w \odot x + b$ with channel-wise
102 parameters $w, b \in \mathbb{R}^C$. BN and LN expose (γ, β) ; ConvNeXt provides layer-scale λ (encoded as
103 $(\lambda, 0)$); ViT applies per-head QKV scaling; NF-Net uses gain g . Treating each pair (w, b) identically
104 permits a single universal adaptor.

105 3.3 Synthetic self-supervision

106 Lacking labels, we adopt entropy minimisation as in Tent, but drive the frozen backbone with $1/f$
107 Gaussian noise and RandAugment distortions. The pre-text task is to reduce softmax entropy after a
108 single correction step, which suffices to train the hyper-network without ever touching real images.

109 3.4 Householder approximation

110 Any orthogonal matrix can be decomposed into a product of Householder reflectors $H_i = I -$
111 $2u_i u_i^\top / \|u_i\|^2$. Truncating after K terms yields an $\mathcal{O}(CrK)$ approximation whose cost scales
112 linearly with K , enabling a graceful latency–accuracy trade-off.

113 3.5 χ^2 -based collapse detection

114 Let ℓ and ℓ' denote logits before and after adaptation. The χ^2 divergence $D_{\chi^2}(\ell' \| \ell) = \sum_i (\ell'_i - \ell_i)^2 / \ell_i$
115 upper-bounds the error increase under arbitrary label shift. A concentration inequality ensures $\mathbb{P} \leq \varepsilon$;
116 threshold $\tau = 3$ keeps $\varepsilon \approx 0.01$ in practice.

Algorithm 1 Meta-training on synthetic noise

```
1: Input: Frozen backbone  $f_0$ , SAUNet parameters  $\theta$ , optimiser, steps  $S$ 
2: for  $s = 1$  to  $S$  do
3:   Sample mini-batch of synthetic images  $X$  (frequency-matched Gaussian + RandAugment)
4:   Compute logits before adaptation:  $\ell \leftarrow f_0(X)$ 
5:   Compute per-layer statistics  $s$  and SAUNet corrections  $(\Delta w, \Delta b) \leftarrow h_\theta(s)$ 
6:   Apply one forward correction to obtain adapted logits  $\ell'$ 
7:   Compute loss  $L \leftarrow E$  ▷ Entropy difference proxy
8:   Update  $\theta$  with Adam step on  $\nabla_\theta L$ 
9: end for
```

117 4 Method

118 4.1 SAUNet layer

119 For each affine pair (w, b) we insert a correction module that outputs $w' = w + \Delta w$ and $b' = b + \Delta b$.
120 During the forward pass we compute per-sample, per-channel standardised activations z and gather
121 mean, variance, skewness and kurtosis into a four-dimensional vector s . A low-rank hyper-network

$$h_\theta(s) = P^\top \sigma(Ts), \quad T \in \mathbb{R}^{4 \times r}, \quad P \in \mathbb{R}^{r \times 2}, \quad r \ll C,$$

122 predicts $(\Delta w, \Delta b)$. σ is tanh. Both matrices are quantised to INT8 offline. With $r = 8$, ConvNeXt-T
123 gains only 0.09 % additional parameters and 1.5 % FLOPs.

124 4.2 Meta-training without source data

125 The backbone is frozen. Each step draws a mini-batch of synthetic images, computes logits before
126 (p_0) and after one SAUNet correction (p_1), and minimises the entropy difference $L = E$. Adam with
127 learning rate 10^{-3} and cosine decay over 25 000 steps converges in roughly two hours on a single
128 A100 GPU.

129 4.3 Any-time inference

130 Upon receiving a latency budget B_t the runtime chooses the smallest K such that measured latency
131 $\leq B_t$. The mapping $\sigma(Ts)$ is approximated by the first K Householder reflectors of its SVD, giving
132 cost $\mathcal{O}(CrK)$. Updates are applied in-place, and a two-tap Welford estimator tracks the magnitude
133 of Δw for long-term stability at 16 bytes per layer.

134 4.4 Collapse guard

135 After adaptation we compute $D_{\chi^2}(\ell' \parallel \ell)$. If the value exceeds τ the update is rolled back and K is
136 doubled; if the bound is still violated the update is discarded. Empirically $\tau = 3$ yields a false-positive
137 rate of 0.4 % and blocks 98 % of actual collapses.

138 4.5 Compatibility

139 SAUNet accesses only activations and writes element-wise scale-bias corrections, making it agnostic
140 to convolutional, transformer and norm-free backbones. Automatic graph traversal wraps all eligible
141 affine pairs (see Supplementary Listing 1).

142 4.6 Pseudocode

143 5 Experimental Setup

144 5.1 Hardware and framework

145 All full-scale experiments run on a single NVIDIA A100-80GB with CUDA 12.2 and PyTorch 2.1;
146 automatic mixed precision uses bfloat16. INT8 kernels reuse the backbone GEMM engine. Three
147 random seeds {17, 42, 94} are used throughout.

Algorithm 2 Any-time inference with χ^2 collapse guard

```
1: Input: Budget  $B_t$ , initial  $K$ , threshold  $\tau$ 
2: Measure base latency; choose smallest  $K$  such that total latency  $\leq B_t$ 
3: Obtain pre-adaptation logits  $\ell$ 
4: Apply SAUNet with Householder- $K$  approximation to get  $\ell'$ 
5: Compute  $d \leftarrow D_{\chi^2}(\ell' || \ell)$ 
6: if  $d > \tau$  then
7:   Roll back update; set  $K \leftarrow 2K$  and re-apply
8:   if  $D_{\chi^2}(\ell'' || \ell) > \tau$  then
9:     Discard update and use  $\ell$  for prediction
10:  else
11:    Accept  $\ell''$ 
12:  end if
13: else
14:   Accept  $\ell'$ 
15: end if
```

148 5.2 Backbones

149 ConvNeXt-Tiny (layer-scale, 28 M parameters), ViT-Small/16 (LN + QKV scale, 22 M), NF-ResNet-
150 50 (norm-free, 25.6 M) and ResNet-50-BN (classic, 25.5 M).

151 5.3 Baselines

152 Source (frozen), BN-stat, TTN (4), UnMix-TNS (5), Tent (1), EATA, CoTTA, UniTTA-0G.

153 5.4 Experiment 1: Breadth-of-coverage corruption

154 ImageNet-C v1.1 (15 corruptions $\times$ 5 severities), Tiny-ImageNet-C, ImageNet-V2, ImageNet-A and
155 DomainNet (Real $\rightarrow$ Clipart/Sketch). Static batch size 64; χ^2 guard disabled. Hyper-network rank
156 $r \in \{4, 8, 16\}$; $r = 8$ selected on Tiny-ImageNet-C severity 3.

157 5.5 Experiment 2: Privacy-preserving meta-training

158 ConvNeXt-T and ViT-S meta-trained under (A) synthetic noise only and (B) 10 % ImageNet-1K.
159 Evaluated on Ego4D object classification and ImageNet-C severity 3. Privacy leakage is quantified
160 via the Yeom membership-inference attack; we report AUROC.

161 5.6 Experiment 3: Streaming video with budgets

162 ImageNet-VID-C and Ego4D streams, batch = 1. An external process issues latency budgets
163 $B \in \{3.3, 6.6, 13\}$ ms every 0.5 s. We compare SAUNet-adaptive (dynamic K), SAUNet-fixed
164 $K = 4$, Tent and the frozen source model. Metrics are mAP@0.5, average latency, budget violations,
165 roll-backs, collapse count and energy per frame.

166 5.7 Common metrics

167 Primary: top-1 error (images) or mAP (video). Secondary: latency, energy per image, parameter and
168 SRAM overhead, and collapse statistics. FLOPs are computed with fvcare; energy via nvidia-smi
169 power integration.

170 6 Results

171 7 Conclusion

172 SAUNet widens the scope of forward-only test-time adaptation beyond normalisation layers, removes
173 the need for source data via synthetic self-supervision, introduces a hardware-friendly any-time

174 compute knob and provides a provable χ^2 safety guard. Across six datasets and four diverse backbones
175 it delivers state-of-the-art robustness at raw-inference speed with less than 0.1 % parameter overhead
176 and a three-percent energy increase. Future work will explore group-wise statistics, extend the
177 framework to dense prediction tasks and analyse the approximation error of Householder truncation
178 under adversarial latency perturbations. We believe SAUNet paves the way for privacy-preserving,
179 low-energy adaptive vision on mobile and safety-critical edge devices without compromising speed
180 or stability.

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